

SOFTWARE AG INTERVIEW EXPERIENCE 2023

About the Company:

Software AG is a German multinational software corporation that develops enterprise software for business process management, integration, and big data analytics. Founded in 1969, the company is headquartered in Darmstadt, Germany, and has offices worldwide.

Total Rounds:

1. Online Assessment
2. Written Coding
3. A) Technical Interview -1
B) Technical Interview -2
4. HR interview

1. Online Assessment

Questions: 3 coding questions

Time: 1 hour

1. Detect and print cycle in an undirected graph
2. Given two arrays, heights and weights, of n persons, where heights[i] represents the height of the ith person and weights[i] represents the weight of the ith person, we want to find the maximum number of persons who can stand one above another.

Solution: Sort the values in increasing order according to one parameter, either height or weight, and then apply the longest increasing subsequence

algorithm to that sorted order to find the solution. You can further optimize this process using dynamic programming (DP).

3. <https://www.geeksforgeeks.org/given-an-array-of-numbers-arrange-the-numbers-to-form-the-biggest-number/>

2. Written Coding:

Venue: CUIC

Questions: 1 coding and 1 debugging

Time: 40 minutes

1. <https://leetcode.com/problems/trapping-rain-water/>

2. Given a code snippet, identify the output for the given input and explain the code.

3.

A) Technical Interview -1

Tell me about yourself

I was asked to choose one of the projects mentioned on my resume

I was told to design that project (System design approach)

Questions after designing:

- What would you do if server is overloaded?
- How do you optimize database performance?
- Can you design microservice architecture?

I was asked a dsa problem

- Write code to allocate N objects to M slots equally
- Write a function to add and remove list of objects from slots
- How would you reallocate such that each slot contains equal number of objects?
- Explain shortest path algorithm in graph

Then, interviewer asked questions from DBMS

- Explain normalization?
- How would you change the data type of existing column in database?

I was asked some questions about the work I have done in my internship

He asked me whether I know full text search in NOSQL (elastic module)

B) Technical Interview -2

The first question the interviewer asked was bit tricky which is, I need to choose a correct data structure for the given requirement. The requirement is similar to hashmap. Let' say

K1 -> v1,v2,v3

K2-> v4,v3

K3-> v6,v4

Sample input: k1 sample output: v1, v2, v3

Sample input: v3 sample output: k1, k2 (fetching keys for the given values)

It can be done using another hashmap by storing all the values as key but you cannot this additional space. So, think of another data structure.

Solution: Graph

The second question was 1 card will be taken from the deck of 52 cards and you will be given remaining 51 cards. You need to find that missing card. Think in a programmatic way. Also, you cannot use additional space. I gave two solutions, interviewer was satisfied with my solution.

I was also asked some managerial questions

- What have you learnt in the college so far?
- How do you keep yourself updated?

4. HR Interview

Tell me about yourself

He asked some questions about my project

I mentioned some ML projects in my resume. So, he asked me some ML concepts.

Some general questions like

- What is your great weakness?
- How is college going?
- How do you keep yourself updated?

Verdict : Selected

TIPS:

Solve DSA problems regularly in Leetcode,HackerRank etc.

Give importance to aptitude as well.

Prepare the projects you have mentioned in your Resume

Prepare well on topics like,

- OOPS
- DBMS
- OS
- System Design (basics)

Prepare for HR questions as well.

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